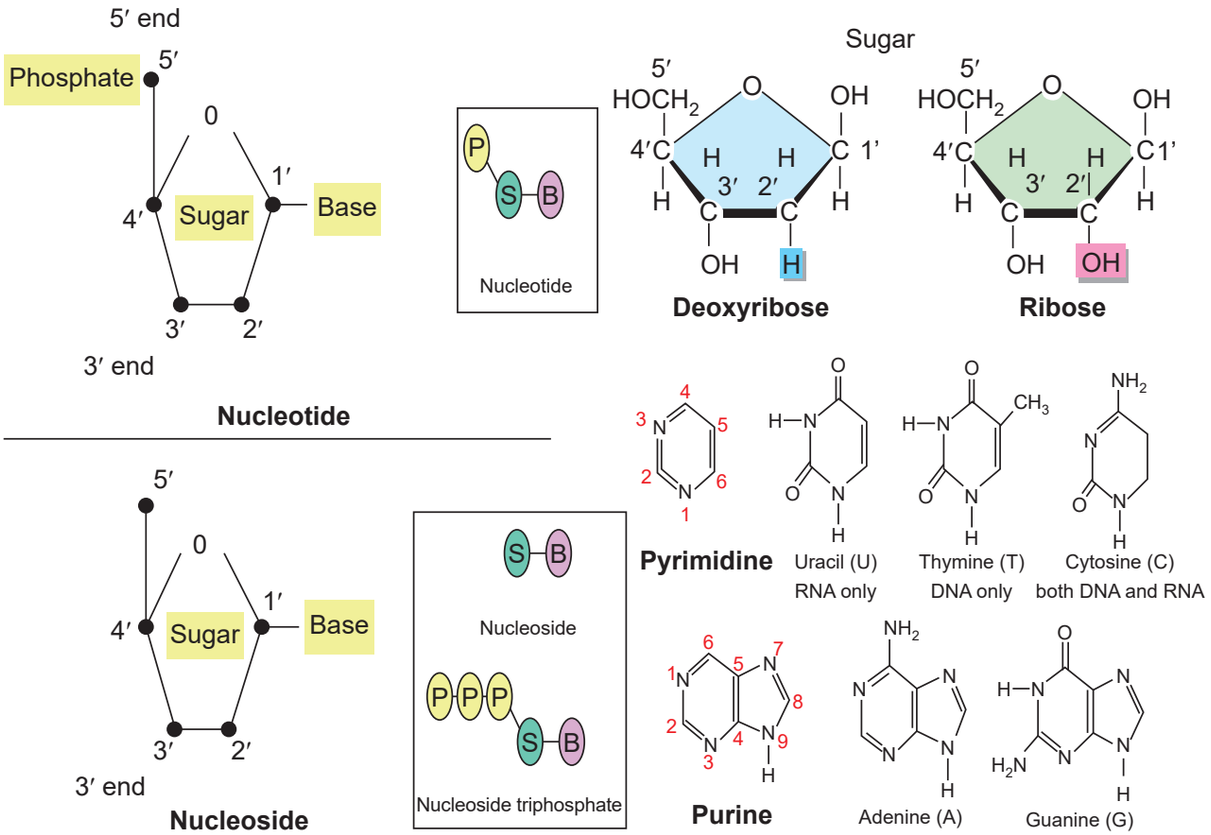


Figure 14.6: Structure of nucleotide, nucleoside, deoxyribose, ribose and nitrogenous bases

- Two polynucleotide chains join together through hydrogen bonds between nitrogenous bases, to form double stranded DNA (Figure 14.7b).
- Two hydrogen bonds exists between adenine and thymine and three hydrogen bonds between guanine and cytosine.
- DNA is coiled in the form of a double helix, in which both the strands of DNA coil around an axis (Figure 14.7d & e).
- The further coiling of this axis upon itself produces DNA supercoiling an important property of DNA structure.

All DNA, whether large or small, possess the same sugar phosphate backbone. What distinguishes one DNA from another is the length of the polymer and distribution of four bases along the backbone. The variety of sequences that can be made from the four nitrogenous bases is limitless, as is the number of melodies possible with a few musical notes. RNA differs from DNA by having a ribose sugar instead of deoxyribose and nitrogenous base Uracil instead of Thymine.

14.2.1 Watson And Crick Model of DNA Double Helix

In the early 1950's, Rosalind Franklin and Maurice Wilkins used the powerful method of X-ray diffraction to shed more light on the structure of DNA. From the X-ray Diffraction pattern it was deduced that DNA molecules are helical. In 1953 Watson and Crick (Figure 14.8) postulated a three dimensional model of DNA structure based on Franklin's X-ray crystallographic studies. In recognition of their work leading to

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